Visual Composition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One current composition record with earlier provenance review preserved.

Name criteria, trace a changed relation, repair a count-only claim and design a new check.

Keep one habit: New answers remain separate from saved provenance responses; keep proposed and observed outputs distinct.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Image Generation Models

Structured Output and Schemas

Prompt as a Contract

W1 · Name composition criteria

> Current tasks use new answer fields; earlier provenance tasks retain saved responses.

Current brief and scope

Main composition brief: exactly two labelled bottles B1,B2 left of one beaker,one arrow bottles→beaker. Original description: bottles right;arrow beaker→bottles. Changed description moves bottles left but retains beaker→bottles. Count/labels pass both;position improves;flow remains wrong.

The native figure reconstructs these descriptions and proposes a full repair; it is not an original generator output or verified chemistry.

Current task and new response

- Explain count, labels, position and semantic arrow endpoints in the main bottle/beaker brief. Give a failure that can remain even when all objects are present.

> **My current visual-composition reasoning**

▼ **Review · Earlier provenance task and saved response**

Reference for this exercise: open this week's student lesson, "Visual Composition", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace a changed relationship

> Current tasks use new answer fields; earlier provenance tasks retain saved responses.

Current brief and scope

Main composition brief: exactly two labelled bottles B1,B2 left of one beaker,one arrow bottles→beaker. Original description: bottles right;arrow beaker→bottles. Changed description moves bottles left but retains beaker→bottles. Count/labels pass both;position improves;flow remains wrong.

The native figure reconstructs these descriptions and proposes a full repair; it is not an original generator output or verified chemistry.

Current task and new response

- For the main original and move-bottles-only descriptions, judge count, bottle labels, left-of position and semantic arrow flow separately. Give the passing evidence and the remaining repair. State whether you inspected an original model image or a classroom reconstruction.

My current visual-composition reasoning

▼ **Review · Earlier provenance task and saved response**

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair a count-only argument

> Current tasks use new answer fields; earlier provenance tasks retain saved responses.

Current brief and scope

Main composition brief: exactly two labelled bottles B1,B2 left of one beaker,one arrow bottles→beaker. Original description: bottles right;arrow beaker→bottles. Changed description moves bottles left but retains beaker→bottles. Count/labels pass both;position improves;flow remains wrong.

The native figure reconstructs these descriptions and proposes a full repair; it is not an original generator output or verified chemistry.

Current task and new response

- The supplied layout has every requested object, but the arrow starts at the wrong object. Does correct object presence establish the required meaning? Explain.

My current visual-composition reasoning

▼ **Review · Earlier provenance task and saved response**

A report omits this qualification: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing composition check

> Current tasks use new answer fields; earlier provenance tasks retain saved responses.

Current brief and scope

Main composition brief: exactly two labelled bottles B1,B2 left of one beaker,one arrow bottles→beaker. Original description: bottles right;arrow beaker→bottles. Changed description moves bottles left but retains beaker→bottles. Count/labels pass both;position improves;flow remains wrong.

The native figure reconstructs these descriptions and proposes a full repair; it is not an original generator output or verified chemistry.

Current task and new response

- Propose a complete new layout description that changes exactly one required relation while retaining the main object counts. State each expected criterion outcome before sketching or checking. Keep the stars/containment independent task separate.

My current visual-composition reasoning

▼ Review · Earlier provenance task and saved response

Reference for this exercise: open this week's student lesson, "Visual Composition", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

W4 · Audit a new spatial description independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Evaluate only the supplied stars/triangle/circle/rectangle description. No original image or measurements are supplied. State which count and relationship checks the description supports and which containment requirement it contradicts; do not invent hidden geometry or inspect the classroom bottle reconstruction as if it were this task's image.

A supplied drawing description says three stars are above a triangle, while a circle crosses the edge of a rectangle. The brief requires three stars above the triangle and the circle fully inside the rectangle.

Evaluate this description.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Create age-fit explanatory visuals with labeled purpose and no invented documentary **evidence**.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Earlier poster/**provenance** completion practice: retain this task and saved response as review. Use the current W2 layout-comparison task for this week's supported practice.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

An image brief requires two trees, a bench between them and the sun upper-right. A result has all objects but the bench is left of both trees. Write a spatial test that detects this.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Brief: large tree left, small bicycle right, open sky above. **Output** puts bicycle left.

Create a visual checklist.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A poster puts pale yellow labels on white and hides the main object behind decorations. Suggest two revisions and how to check them.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 11 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

An image brief requires two trees, a bench between them and the sun upper-right. A result has all objects but the bench is left of both trees. Write a spatial test that detects this.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Earlier two-arrow provenance audit

REFERENCE · SECTION 15

Additional practice · Earlier two-arrow provenance audit

> This retained task is separate from the selected stars/containment assessment.

Earlier task and saved response

- A generated diagram requires two arrows and labels **input**/**output**. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental **observation**. Audit.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

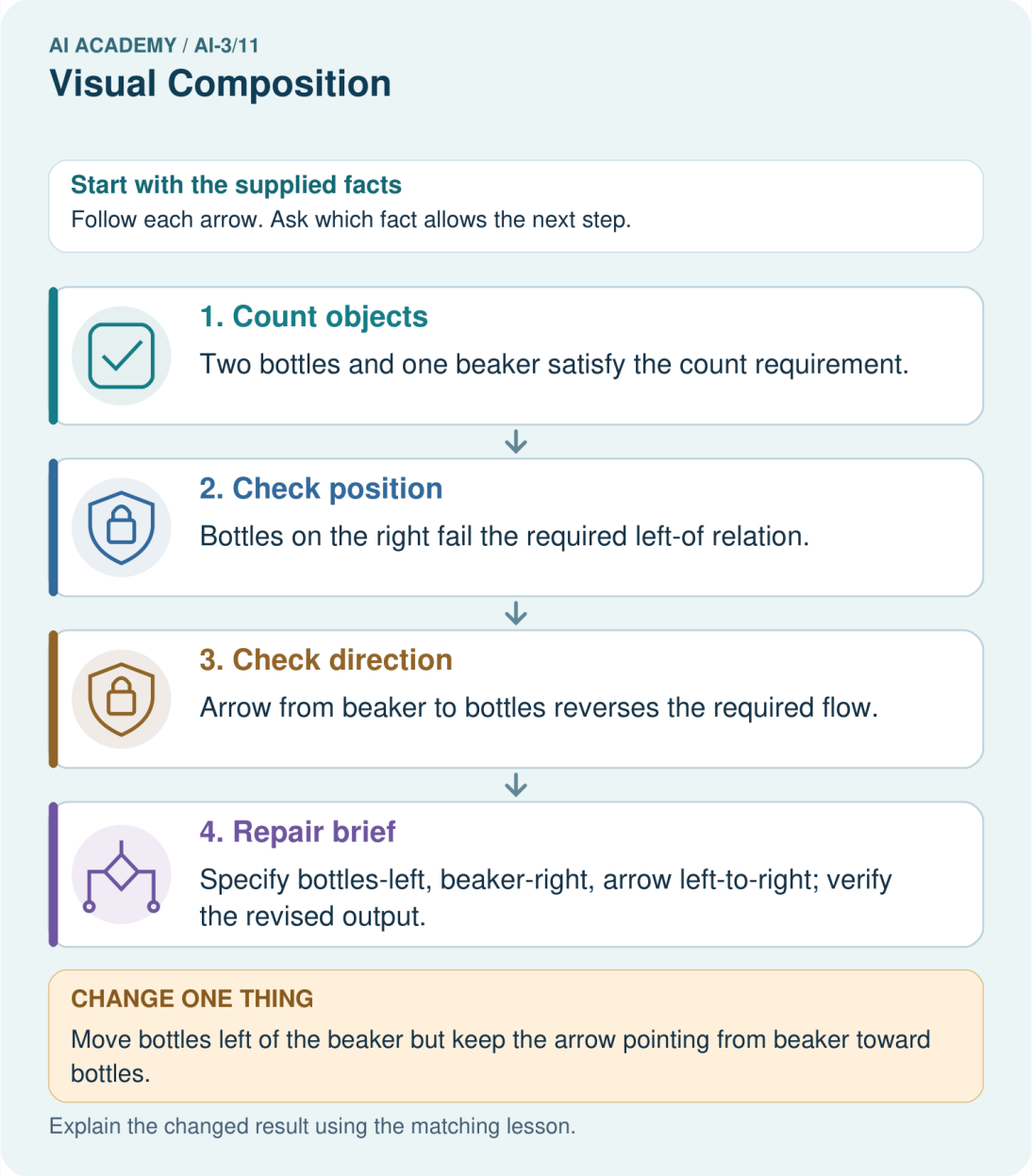
> Keep the explanation and the support you used.

Use this when helpful

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.

Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which checks pass after that single change, and what still needs repair?

> **My picture-based explanation and changed-case prediction**

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.
- Which labelled step matches this detail? Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.
- 1. Check position

- 2. Repair brief
- 3. Count objects
- 4. Check direction
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Repair brief Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.

Having every requested object is not the same as conveying the requested **relationship**. Position and arrow direction can change a scientific explanation while object counts remain correct.

Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Changed-case question

- Changed-case practice: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Position now passes; arrow direction still fails.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Position now passes; arrow direction still fails. Changing object placement does not automatically reverse an arrow's semantic source and destination.

Repair and save your work

Review Image Generation Models (ai-3/10). Goal: Explain the **mechanism** and terminology of image generation and **provenance**.

Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Review Prompt as a Contract (ai-3/5). Goal: Write a prompt specification with observable response criteria. Short practice: Why should the tool not fill melting from memory even if it knows a common definition?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Visual Composition: the case

AI ACADEMY / AI-3/11 · 01 · LOOK AT THE FACTS

Visual Composition: the case

The situation

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

Supplied evidence

Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.
Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.

By the end, I can check object counts and spatial relations as separate visual requirements.

Follow the labelled picture with the matching explanation. Open the vector diagram

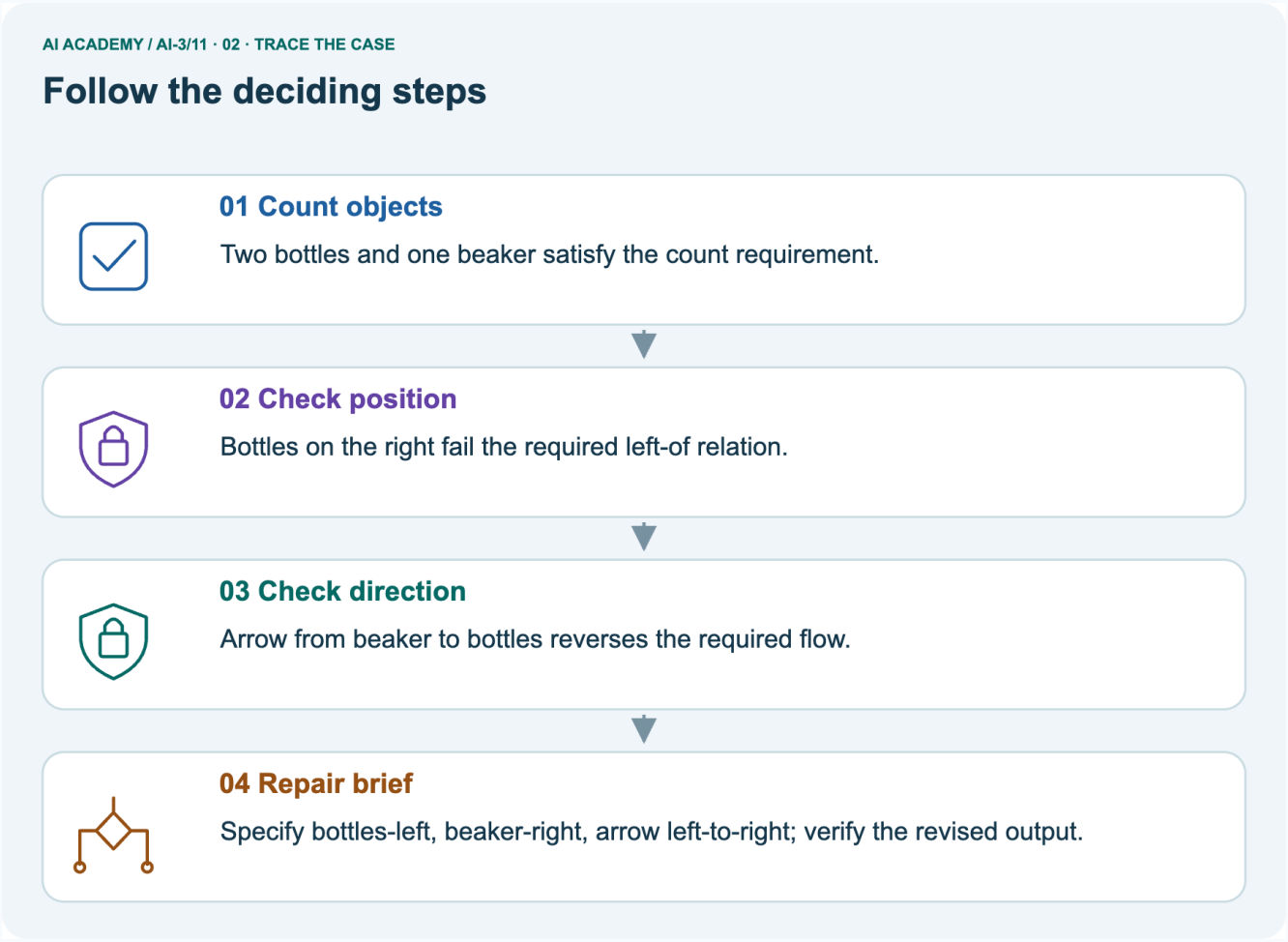
A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description. Supplied facts: Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.

Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



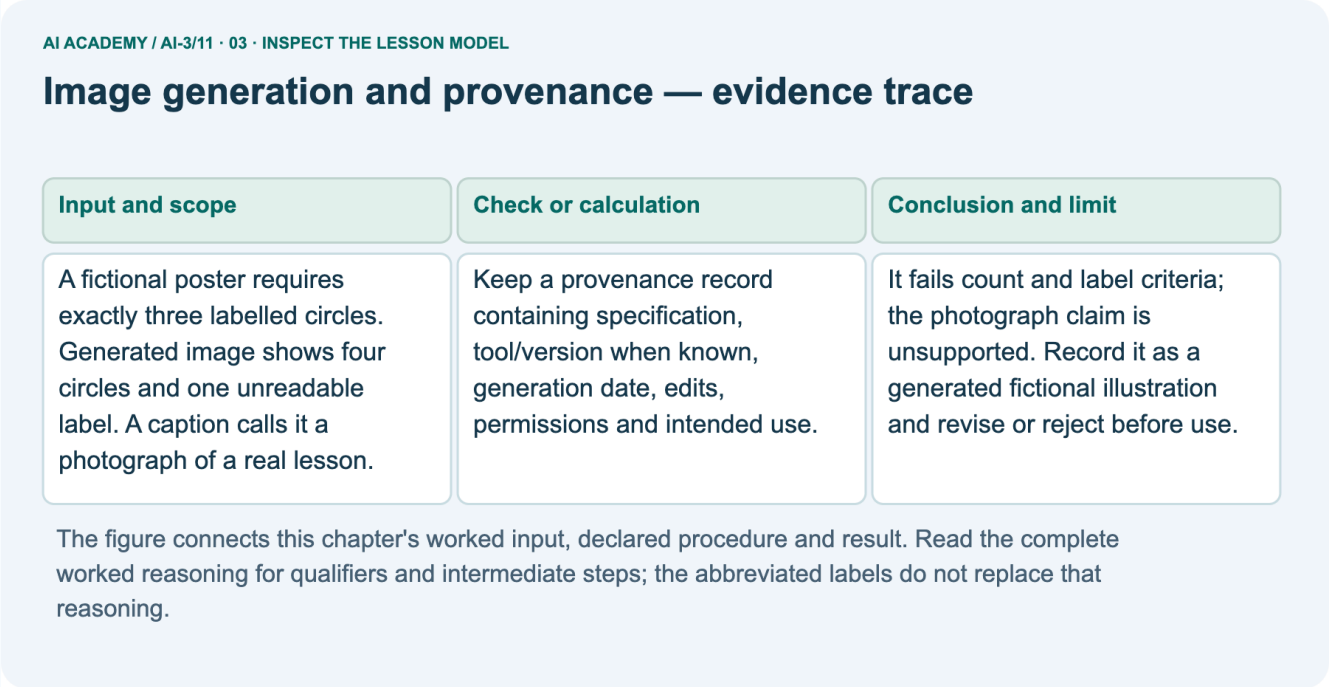
Follow the labelled picture with the matching explanation. Open the vector diagram

Count objects: Two bottles and one beaker satisfy the count requirement. → Check position: Bottles on the right fail the required left-of relation. → Check direction: Arrow from beaker to bottles reverses the required flow. → Repair brief: Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Image generation and provenance — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Image generation and **provenance** — **evidence** trace. A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label.

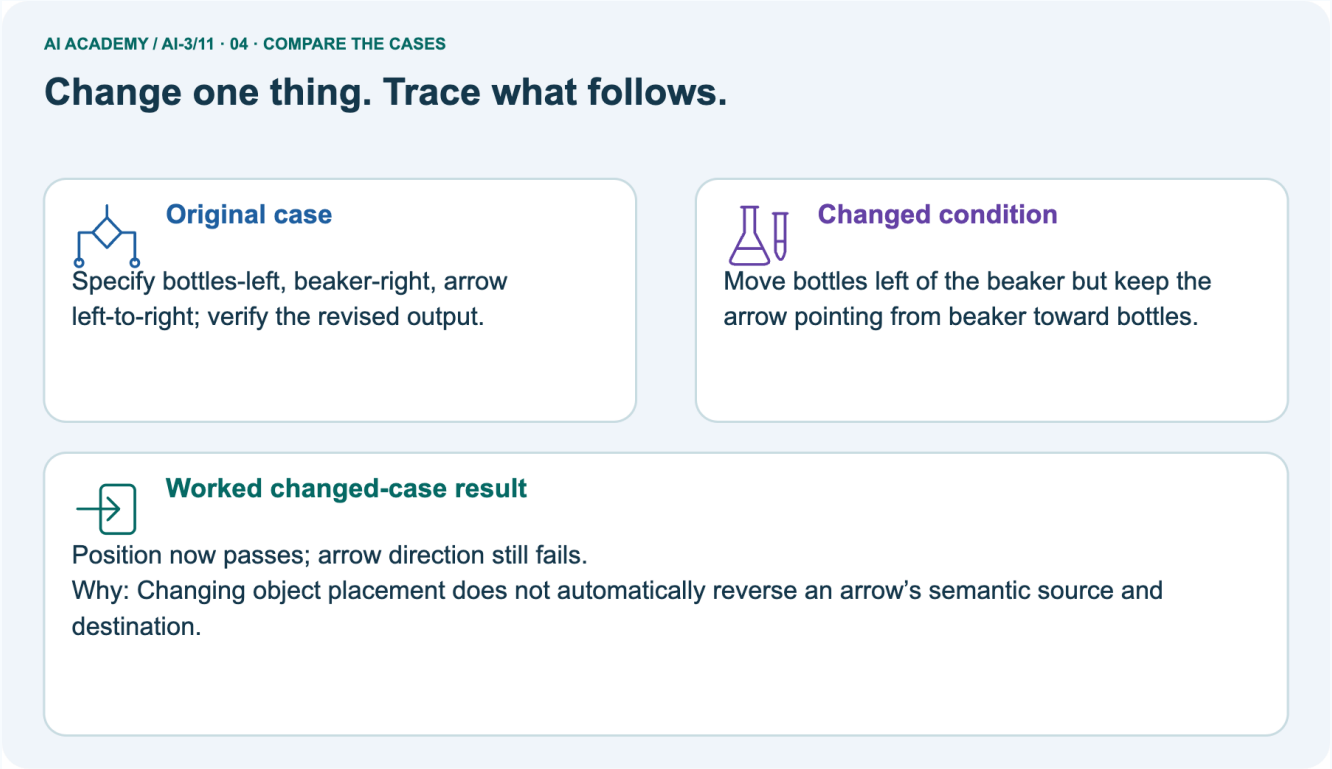
A caption calls it a photograph of a real lesson. | Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. | It fails count and label criteria; the photograph claim is unsupported.

Record it as a generated fictional illustration and revise or reject before use.

Read the labels and values. Explain the **relationship** the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

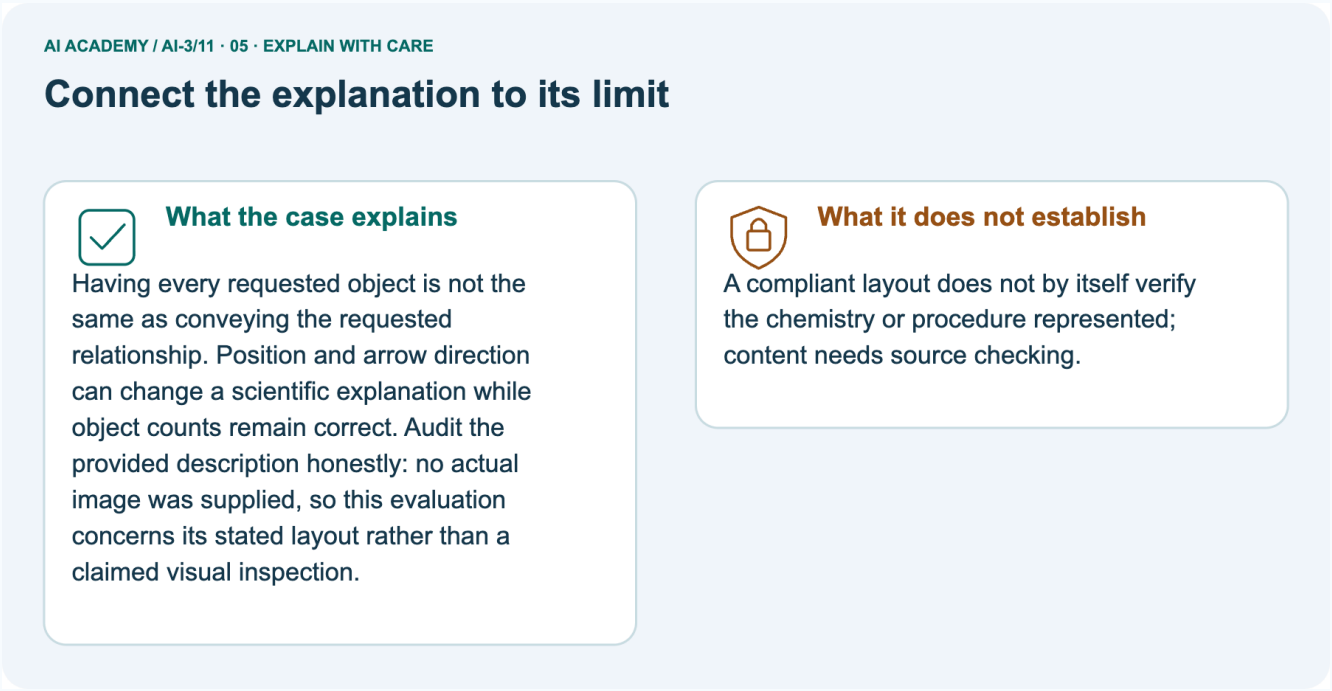
Original case: Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**. Changed condition: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

Expected result: Position now passes; arrow direction still fails. Reason: Changing object placement does not automatically reverse an arrow's semantic source and destination.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Having every requested object is not the same as conveying the requested **relationship**. Position and arrow direction can change a scientific explanation while object counts remain correct.

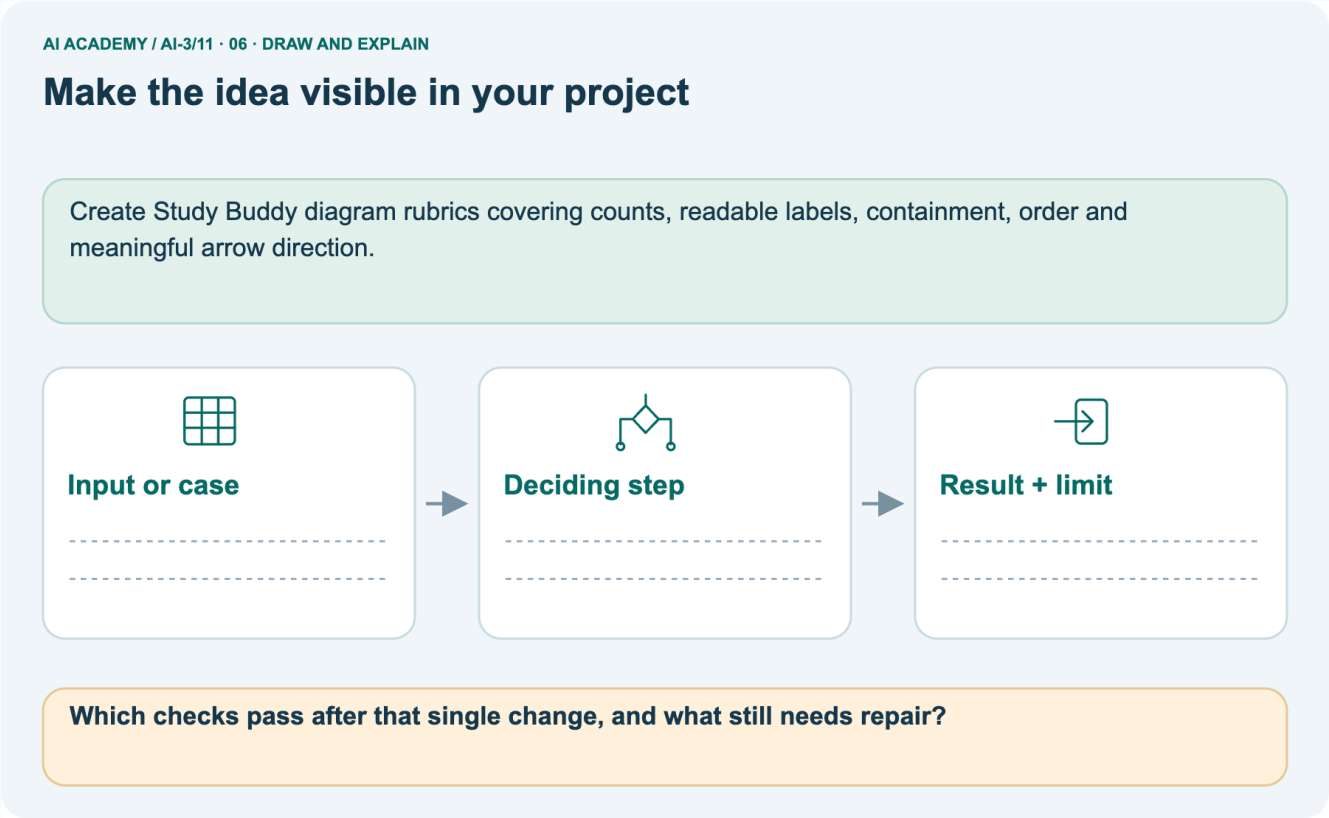
Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Limit: A compliant layout does not by itself verify the chemistry or procedure represented; content needs source checking.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction. Draw your **input**, deciding step, result and limit.

Supported question: Which checks pass after that single change, and what still needs repair?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Visual Composition: the case

02. Follow the deciding steps

03. Image generation and provenance — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.